

4-Bit Processor Design and Simulation using Verilog HDL

Overview

This project presents the design and simulation of a simple 4-bit processor implemented using Verilog HDL. The processor demonstrates fundamental CPU concepts including instruction execution, arithmetic and logical operations, register management, memory access, and control logic.

Features

- Custom 4-bit Processor Architecture
- 4-bit Arithmetic Logic Unit (ALU)
- Register File with 4 General Purpose Registers
- Control Unit for Instruction Decoding
- Data Memory Module
- Custom Instruction Set Architecture (ISA)
- Verilog HDL Implementation
- Testbench for Functional Verification
- Waveform-Based Simulation Analysis

Processor Architecture

Processor Architecture

Instruction Format

Bits	Description
7:4	Opcode
3:2	Destination Register (Rd)
1:0	Source Register (Rs)

Register Set

Register	Description
R0	General Purpose Register
R1	General Purpose Register
R2	General Purpose Register
R3	General Purpose Register

Instruction Set

Opcode	Mnemonic	Operation
0000	NOP	No Operation
0001	ADD	$Rd = Rd + Rs$
0010	SUB	$Rd = Rd - Rs$
0011	AND	$Rd = Rd \& Rs$
0100	OR	$Rd = Rd \mid Rs$
0101	XOR	$Rd = Rd \wedge Rs$
0110	MOV	$Rd = Rs$
0111	LOAD	$Rd = MEM[Rs]$
1000	STORE	$MEM[Rs] = Rd$
1001	JMP	Jump

Project Structure

4bit-processor/

├── src/

| ├── ALU.v

| ├── RegisterFile.v

| └── DataMemory.v

```
| |—— ControlUnit.v
| |—— Processor4bit.v
|
|—— testbench/
| |—— tb_processor.v
|
|—— waveforms/
| |—— cpu_waveform.png
|
|—— docs/
| |—— Processor_Architecture.png
|
|—— README.md
|—— LICENSE
|—— .gitignore
```

Simulation

The design was verified using Verilog simulation tools such as:

- ModelSim
- Vivado Simulator
- GTKWave

Simulation Flow

1. Compile Verilog modules
2. Compile testbench
3. Run simulation
4. Generate VCD waveform file

5. Analyze results using GTKWave

Sample Execution

Initial Values:

R1 = 5 (0101)

R2 = 3 (0011)

ADD R1, R2

$0101 + 0011 = 1000$

Result:

R1 = 8

SUB R1, R2

$0101 - 0011 = 0010$

Result:

R1 = 2

Waveform Results

Simulation Waveform

Future Enhancements

- Zero Flag
 - Carry Flag
 - Overflow Detection
 - Conditional Branch Instructions
 - Immediate Operations
 - Stack Pointer
 - Pipelined Architecture
 - FPGA Implementation
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Applications

- Digital System Design Learning
 - Computer Architecture Fundamentals
 - Verilog HDL Practice
 - FPGA Development Preparation
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Author

Your Name

B.Tech Electronics and Communication Engineering

Verilog HDL | Digital Design | VLSI

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